

Effect of Airfoil-Preserved Undulations on Wing Performance

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Most leading-edge tubercles studies involve serrated-type or undulated leading edges where the airfoil cross-section is destroyed unintentionally. Experimental investigations were performed at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) on an undulated wing where the NACA 0012 airfoil cross-section is preserved along the wingspan. Sensitivity study was done on the number of undulations along the span (6, 9 and 12) and undulation placement (leading edge, trailing edge, and both leading and trailing edge). The effect of the undulations on the wingtip vortex was also investigated through Particle Image Velocimetry (PIV). The wing with trailing edge undulations gave better aerodynamic performance and lower parasitic drag when compared to the baseline and other wing cases at higher angles of attack. The aerodynamic coefficients were found to be independent of the number of undulations. For the same coefficient of lift, the undulated wing cases reduced the wingtip vortex circulation by 25% which in turn affected the balance of induced and parasitic drag. Investigation into the wingtip vortex RMS and aerodynamic efficiency revealed an inverse relationship.

Nomenclature

α	= Angle of attack, degrees	D	= Drag force, N
λ	= Wavelength, cm	e	= Span efficiency
$\Gamma(r)$	= Circulation as a function of radius, m ² /s	L	= Lift force, N
Γ_0	= Total circulation, m ² /s	R	= Strain rate, 1/s
μ	= Dynamic viscosity, Pa-s	r	= Vortex radius, m
ρ	= Density, kg/m ³	r_c	= Vortex core radius, m
σ	= Vortex wandering amplitude, m	U	= X-component of velocity, m/s
σ_x	= RMS wandering x component, m	U_{RMS}	= RMS, m/s
σ_y	= RMS wandering y component, m	u'	= X-component Fluctuating velocity, m/s
Ω	= Vorticity, 1/s	AR	= Aspect ratio
A	= Amplitude, cm	LE	= Leading edge
C_D	= Coefficient of drag	LEU	= Leading edge undulations
C_{Di}	= Coefficient of induced drag	$LETEU$	= Leading edge trailing edge undulations
C_{DP}	= Coefficient of parasite drag	TE	= Trailing edge
C_L	= Coefficient of lift	TEU	= Trailing edge undulations
$C_{L,\alpha}$	= Lift curve slope		

I. Introduction

Even with major advancements in aviation technology, an effective reduction of induced drag has remained a velleity for many years. The lift-induced drag is responsible for more than 70% of the total drag of the aircraft during take-off and landing and about 5-15% of the total drag during cruising [1]. So far, no universal mathematical relation exists which relates the physics and properties of the wingtip vortex roll-up process, its evolution and the induced drag. Over the years, numerous methods to reduce induced drag have been conceived and implemented. Methods such as installing end plates at wingtips, winglets (most common), lift distribution tailoring (by changing the deflection of flaps on the wing), active/passive flow control methods (blowing or suction of air at the wingtip), etc. have been employed to affect the aerodynamic efficiency of the wing. Of all the methods mentioned above, the total reduction in drag so far has only been 5-7%. The non-satisfactory performance of these methods is due to the insignificant influence by the devices on the wingtip vortex roll-up process. At high Reynolds numbers, the wingtip vortex core

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laminarizes irrespective of any external influences. This phenomenon is quantified by the Richardson number defined in Holzapfel et al. [2] which evaluates the stratification of the vortex by evaluating the swirl velocity gradients in the vortex flow. Cotel and Breidenthal [3] and Cotel [4] determined a threshold value of Richardson number given by a function of vortex Reynolds number. For a higher “persistence parameter”, defined by the ratio of rotational to translational speed of the vortex, the threshold value of the Richardson number was experimentally found to be $Re_{Vortex}^{1/4}$. Only laminar flow is possible in the radial location of the vortex containing Richardson number greater than this threshold value. Any perturbations will either be relaminarized or suppressed and any diffusion in the vortex core is only possible at a molecular level [5]. If the threshold value of the Richardson number occurs at a smaller radial location from the vortex core then the majority of the wingtip vortex is turbulent, yielding a poor response to any flow control methods employed at the wingtip. Recent studies have shown that the strength of the wingtip vortex can be significantly affected by enhancing the free shear layer interaction with the wingtip vortex [6], [7], [8]. The blockage of the spanwise flow along the surface of the wing using active and passive flow control methods such as stall fences, dimples, bumps, riblets, etc. is shown to have a strong influence on the growth and evolution of the wingtip vortex as well [9], [10], providing motivation for this study.

Historically, investigations into the leading edge protrusions and surface features on wings was primarily bioinspired. Wings with unusual contours and undulations appear many places in nature, including the humpback whale, bat, and birds (Fig. 1). The humpback whale flipper has tubercles along the leading edge and on the trailing edge towards the tip of the flipper. The whale’s ability to perform tight maneuvers is primarily attributed to the superior performance caused by the presence of these tubercles. Apart from whales, wings with surface features can also be found on bats and birds. Bat wings are naturally contoured as the skin stretches across the bones which is comparable to how canvas stretches across the wooden ribs of a wing to provide a lifting surface. The feathers of a bird’s wing add the necessary surface roughness to increase aerodynamic performance. Deriving similar motivation from wings found in nature, the main focus of this research will be to investigate the changes in aerodynamic performance of wings with leading and trailing edge undulations, the effect of the undulations on the wingtip vortex, and the balance of lift induced and parasitic drag. This combination of these features was achieved by preserving the thickness to chord ratio of the airfoil along the span of the airfoil while changing the chord as a function of span.



Fig. 1 Contoured wings found in nature, (Adopted from [11], [12], [13])

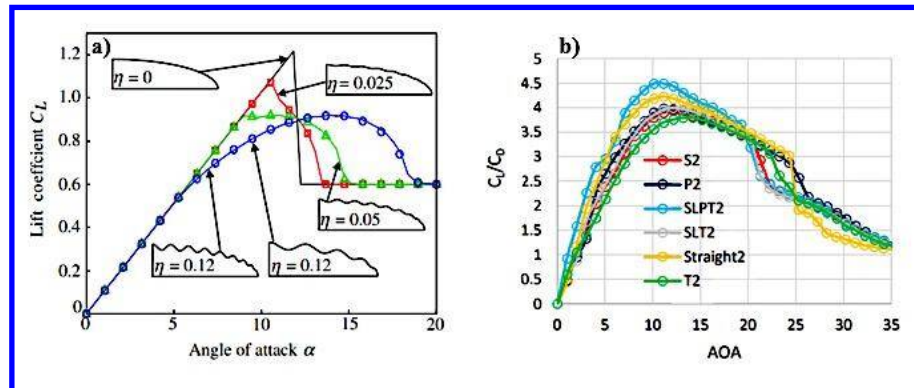


Fig. 2 a) C_L versus angle of attack for baseline wing and tubercle wings of varying amplitude (Adopted from [14]) b) C_L/C_D versus angle of attack for wings with altered leading edges/trailing edges (Adopted from [15])

Van Nierop et al. [14] investigated wing models with leading edge tubercles and found that the tubercles alter the pressure distribution resulting in boundary layer separation behind the tubercles, ultimately leading to a more gradual onset of stall and a higher stall angle (Fig. 2a). Results from [14] indicate that the C_L for the wing with the highest number of tubercles begins to deviate from the linear trend starting at 5° and has a significantly higher stall angle when compared to the baseline. This shows the feasibility of using wings with leading edge tubercles which provides greater stall performance without having a dramatic decrease in lift.

Shorbagy et al. [15] also performed a low-Reynolds number study on wings with leading edge tubercles and found that there was an increase in C_L/C_D for wings with sinusoidal leading edge tubercles and trailing edge tubercles at the wingtip. Figure 2b shows the results from this study for an AR 2 wing at $Re=174,000$. In Fig. 2b, the SLPT2 line that has the highest performance corresponds to a wing with a full sinusoidal leading edge and a straight trailing edge until 56% along the span where the remainder is sinusoidal. This is the only wing that was found to outperform the baseline between 8 and 15° .

Farouk et al. [16] carried out a computational study with spherical protrusions on the leading edge of a NACA 0012 wing as shown in Fig. 3a. Even though the results indicate lower lift at lower angles of attack, the post-stall performance of a wing of this type is found to be more favorable than the baseline (Fig. 3b). The lift curves also show a slight sensitivity to changes in Reynolds number. Based on the above-mentioned literature studies, it is expected that the wings with leading edge undulations will begin to stall earlier than the baseline wing but have a more gradual stall.

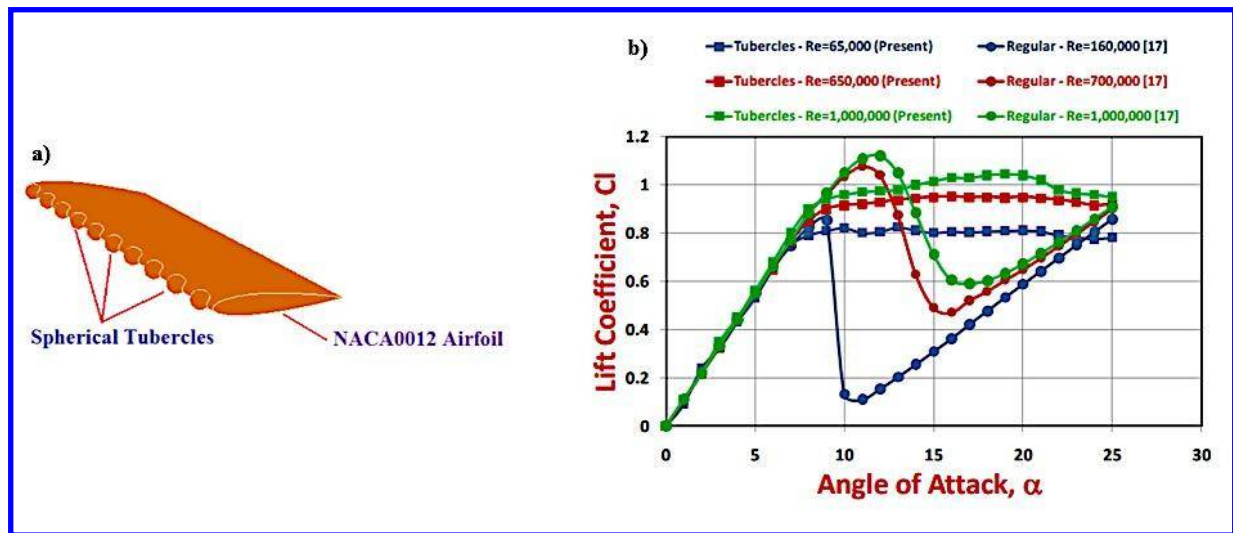


Fig. 3 a) Wings with spherical tubercles b) Effect of spherical tubercles on C_L (Adopted from [16])

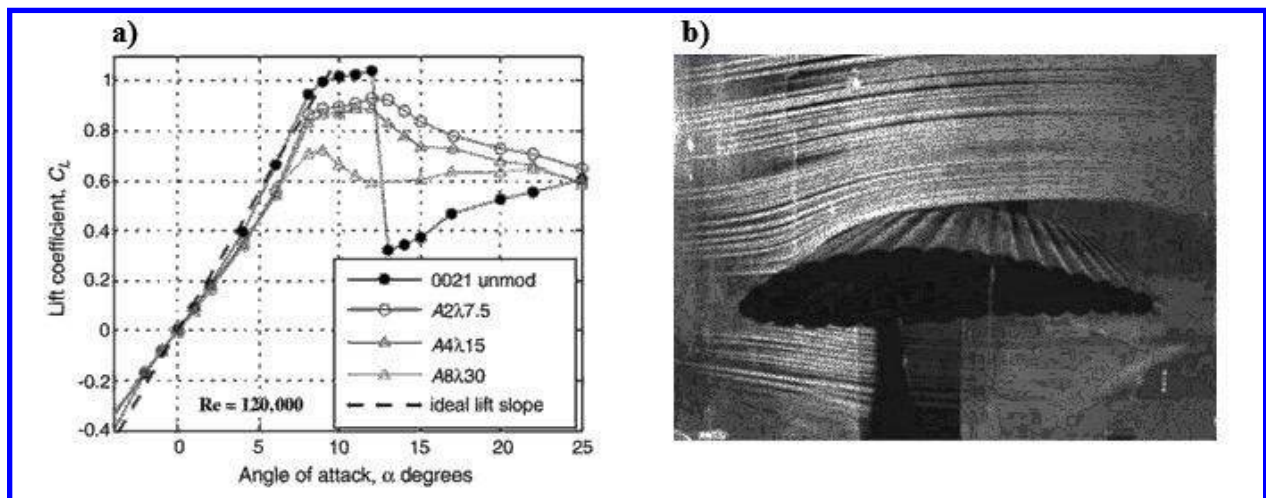


Fig. 4 a) Tubercle amplitude/wavelength ratio and the effect on C_L for NACA 0021 (Adopted from [17]) b) Flow around an inflatable airfoil (Adopted from [18])

Hansen et. al [17] investigated the changes in wing performance with leading edge tubercles of varying wavelength and amplitude for distinct airfoil profiles and found that the tubercle parameters do not make a major impact in the pre-stall angle of attack, but there is a very noticeable impact in the post-stall range as shown in Fig. 4a.

Unintentional surface undulations can also be seen on inflatable wings as shown in Fig. 4b. Zhang et al. [18] performed an analysis of an inflatable airfoil and found that the bumps along the surface reduce flow separation along the airfoil and a similar post-stall performance seen Figs. 2 and 3. Therefore, any changes in surface roughness either spanwise or chordwise results in better post-stall performance while having negligible impact at lower angles of attack.

As this section evinces, vast amounts of literature exist on the effect of leading edge tubercles on wing performance [14], [15], [16], [17], [19], and stall delay [20], [21] and has been considered for applications such as wind turbines [22] and propellers [23]. However, one major difference between the previous studies mentioned above and the present is that in all the prior literature, the airfoil cross-section is destroyed along the wingspan when the leading or trailing edge tubercles are incorporated into the wing design. Preserving the airfoil along the span while creating the undulations is hypothesized to provide better aerodynamic performance and predictability at both pre and post-stall angles of attack. It is also hypothesized that the undulations created by preserving the airfoil will change the balance of induced and parasite drag of the wing.

With the advent of 3D printing and high-speed computing technology, the leading edge undulated wings have recently regained focus through both experimental and computational investigations. While prior research provides ample evidence to the benefits of the undulations, it falls short of providing insight into the way the tubercles, undulations and surface features affect the balance of induced and parasite drag of the wing. Therefore, the objectives of the present research are to elucidate the changes in pre-stall and post-stall wing performance between the airfoil preserved and non-airfoil preserved wings (found in literature) with surface undulations. Sensitivity analysis on the number and placement of the surface undulations was also conducted by comparing the changes in aerodynamic coefficients and efficiencies between different cases. In order to gain insight into the changes in balance of induced and parasite drag due to the surface undulations, the near wake free shear layer wake structure and the wingtip vortex roll-up was investigated as well. However, only the wingtip vortex properties and the aerodynamic coefficients are discussed in this paper. Preserving the airfoil cross-section along the wing allows for development of a theoretical framework to predict lift and drag coefficients as a function of surface undulations which will be discussed in the future publications along with the changes in free shear layer wake.

II. Experimental Setup

A. Wind Tunnel

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open jet configuration (Fig. 5). The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens and 4 interchangeable 76.2cm x 76.2cm x 243.8cm (30" x 30" x 96") test sections. A photo of the UD-LSWT open jet configuration is shown in Fig. 5.

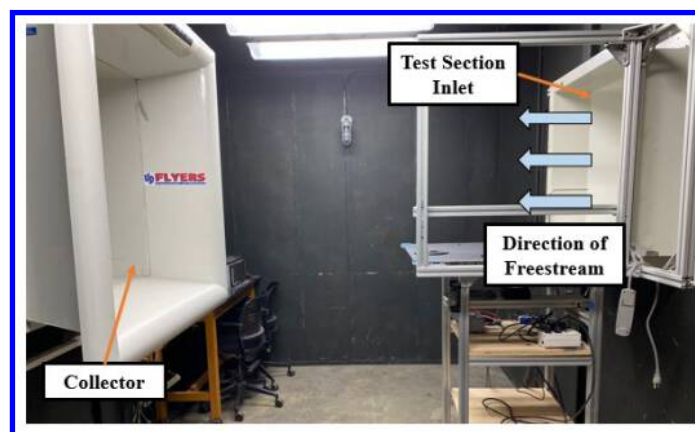


Fig. 5 UD-LSWT in open-jet configuration

The test section is convertible from a closed-jet configuration to an open-jet configuration with the freestream range of 6.7 m/s (20 ft/s) to 40 m/s (140 ft/s) at a freestream turbulence intensity below 0.1% at 15 m/s measured by hot-wire anemometer. The contraction feeds into a pressure/air sealed room where the test section is located. The effective

length of the test section in the open jet configuration is 182 cm (72"). A 137 cm x 137 cm (44" x 44") collector collects the expanded air on its return to the diffuser. The velocity variation for a given RPM of the wind tunnel fan is found using a Pitot tube connected to an TSI T600 Micromanometer.

B. Test Model Design and Fabrication

Nine different airfoil-preserved undulated wing cases (3 with trailing edge undulations, 3 with leading edge undulations and 3 with both leading and trailing edge undulations) were considered for the sensitivity study. The three cases for each undulation placement have different number of undulations on the wing surface (6, 9 and 12). The different cases are depicted in Fig. 6. The baseline wing has a chord length of 12.7 cm and a wingspan of 25.4 cm resulting in a semi-span AR of 2 (full span AR 4). The undulated wings have a mean chord of 11.43 cm (4.5 in) and an effective full-span AR of 4.5. Table 1 depicts the wavelength, amplitude, surface area, and planform measurements for each undulated wing. A NACA 0012 airfoil was maintained throughout the span for all wing cases. The planform area was kept constant for all undulated wing at 285.35 cm² and the baseline NACA 0012 wing planform area was 325.48 cm².

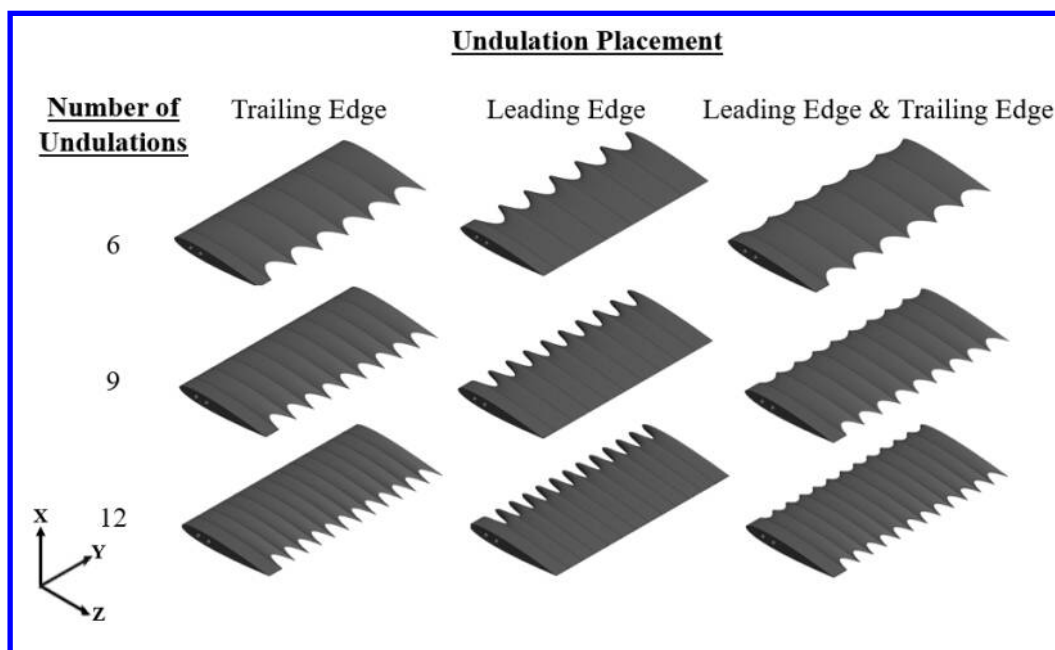


Fig. 6 Schematic of wing models with varying undulation numbers and locations

Table 1 Dimensions for undulated wings

Front View		Case	A, cm	λ , cm	Surface Area, cm ²	Planform Area, cm ²
	Wavelength, λ	6 LE	2.54	3.91	313.06	285.35
		6 TE	2.54	3.91	308.23	
		6 LE + TE	0.635 (LE) 1.905 (TE)	3.91	308.02	
	Amplitude, A	9 LE	2.54	2.61	319.95	
	Wavelength, λ	9 TE	2.54	2.61	311.16	
		9 LE + TE	0.635 (LE) 1.905 (TE)	2.61	310.71	
		12 LE	2.54	1.96	328.00	
		12 TE	2.54	1.96	315.17	
		12 LE + TE	0.635 (LE) 1.905 (TE)	1.96	314.32	
	12.7 cm.					
	25.4 cm.					

All the test models shown in Fig. 6 were generated in SolidWorks. To generate the airfoil preserved undulations, two NACA 0012 airfoil profiles with different chord lengths were used. The larger airfoil had a chord of 12.70 cm while the smaller airfoil had a chord length 10.16 cm. The profiles were placed a certain distance from each other depending on the desired number of undulations, and then were lofted. In the case of wings with trailing edge undulations (TEU), the leading edge of both the smaller and larger airfoil profiles were kept at a constant location. For the wing with leading edge undulations (LEU), the trailing edge of both airfoils were kept at a constant location. The wings with leading edge and trailing edge undulations (LETEU) were modeled by placing the aerodynamic center of both airfoil profiles at the quarter-chord of the entire wing. The mean quarter-chord of the airfoils on TEU wings is at 22.5% of the main wing chord and for the LEU wings, it is at 27.5% of the main wing chord. The loft command allows the airfoil cross-section to be preserved in all wings. The loft was then mirrored to produce the desired number of undulations. All wings were 3D printed using the University of Dayton Gorilla Maker 3D printer.

C. Force-Based Experiment

The ATI Gamma Sensor was used to measure the normal force (N) and axial force (A) on the wing which were transformed into lift and drag forces which were later converted to lift and drag coefficients, C_L and C_D . The range and resolution for each direction of the sensor is shown in Table 2. The sensor was located underneath the splitter plate along with the Griffin motion rotary stage which was used to change the angle of attack. The rotary stage was controlled using Galil motion software. A schematic of force-based test setup is shown in Fig. 7.

Table 2 Specifications of ATI Gamma Sensor

	F_x , N	F_y , N	F_z , N	T_x , Nm	T_y , Nm	T_z , Nm
Range	65	65	200	5	5	5
Resolution	1/80	1/80	1/40	10/13333	10/13333	10/13333

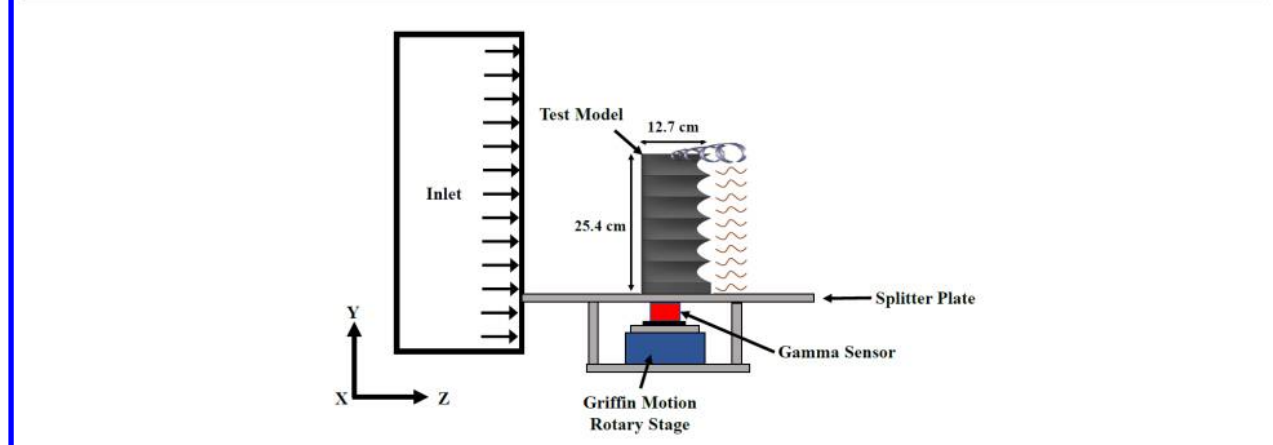


Fig. 7 Schematic of force-based experiments and equipment used in UD-LSWT

Table 3 Test Matrix for Force-Based Experiments

Case	Number of Undulations	Undulation Placement	Wing size Chord x Span (m)	Freestream speed, m/s	AOA Range, Deg	AOA Increment, Deg	Reynolds Number
1	6	LE	0.127 x 0.254	35	-17 to 17	1	282,000
2	9						
3	12						
4	6	TE					
5	9						
6	12						
7	6	LE + TE					
8	9						
9	12						
10	0	N/A					

Table 3 shows the test matrix for the force-based experiments. All undulated wings and the baseline NACA 0012 wing were tested at a freestream velocity of 35 m/s which corresponds to a Reynolds number of approximately 282,000. The experiments were conducted at an angle of attack range of -17° to 17° in 1° increments. Two trials of each case were conducted to ensure repeatability.

D. PIV Experiment

Cross-stream Particle Image Velocimetry (PIV) was conducted to analyze how the undulations affected the wingtip vortex compared to the straight NACA 0012 wing (Fig. 8). The PIV experiment was conducted using a Vicount smoke seeder with glycerin oil, a 200 mJ/pulse Nd:YAG frequency doubled laser (Quantel Twins CFR 300) and an Imperx B2021 camera with 200 mm lens. A plano-convex lens and a plano-concave lens were used in series to open the laser beam into a sheet. The laser sheet was placed 3 chord lengths behind the wing. The laser and camera were triggered simultaneously using a Quantum composer pulse generator. In each test case, 500 image pairs were obtained and processed using ISSI Digital Particle Image Velocimetry (DPIV) software. Two iterations were performed in DPIV processing with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. Table 4 shows the test matrix for the cross-stream PIV experiment. The cross-stream PIV experiment was conducted at a Reynolds number of 201,000. This Reynolds number is lower than the Reynolds number observed for the force-based experiments in Table 2. This is because the freestream speed for the PIV experiment was chosen to be 25 m/s rather than 35 m/s due to wingtip vortex wandering being more prevalent at higher Reynolds numbers. At a lower Reynolds number, the changes in the vortex wandering between the wings can be quantified accurately. The camera was located more than 10 chord lengths downstream from the trailing edge of the wing in an effort to reduce the effects of the camera disrupting the flow. The time delay between the laser pulses was changed at each angle of attack to obtain 8 to 10-pixel particle displacement at the wingtip vortex core boundary.

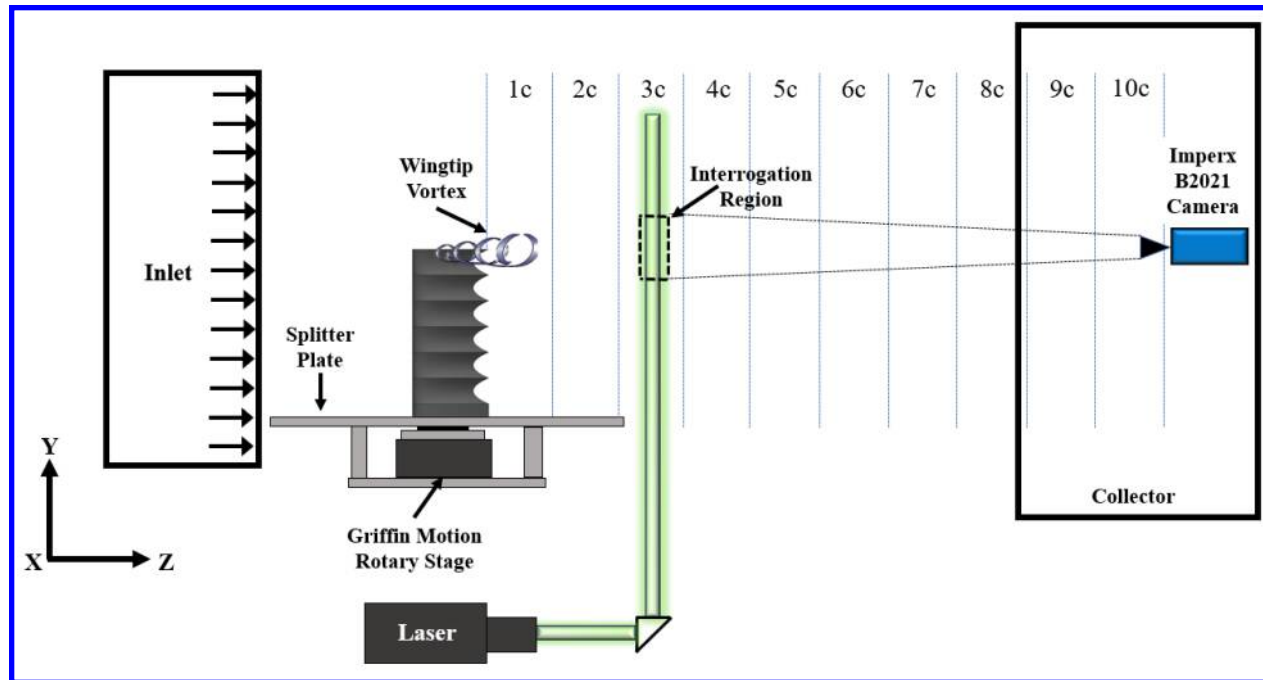


Fig. 8 Schematic of PIV experiments and equipment used in UD-LSWT

Table 4 Test Matrix for Cross-Stream PIV Experiment

Case	Number of Undulations	Undulation Placement	Size, m x m	Freestream speed, m/s	AOA Range, °	AOA Increment, °	Reynolds Number
1	12	LEU	0.127 x 0.254	25	2 to 10	2	201,000
2	6	TEU					
3	12	LETEU					
4	0	N/A					

III. Force-Based Experiment Results

A. Coefficient of Lift

The coefficient of lift results from the force-based experiments are shown in Fig. 9 below. The lift coefficient variation of all the wing cases are compared with the ideal lift curve slope predicted by the Helmbold Equation shown in Eq. (1):

$$C_{L,\alpha} = \frac{\pi AR}{\left[1 + \sqrt{1 + \left(\frac{AR}{2}\right)^2}\right]} \quad (1)$$

It can be seen in Fig. 9 that the lift coefficient for all the cases remains linear at lower angles of attack and agrees extremely well with theoretical predictions for AR 4 and AR 4.5. Major differences in the lift coefficients are only seen at higher angle of attack cases. The increased roughness of the wing due to surface undulations did not alter the linearity of the lift coefficient as a function of angle of attack. This could primarily be due to the preservation of the airfoil profile along the span of the wing.

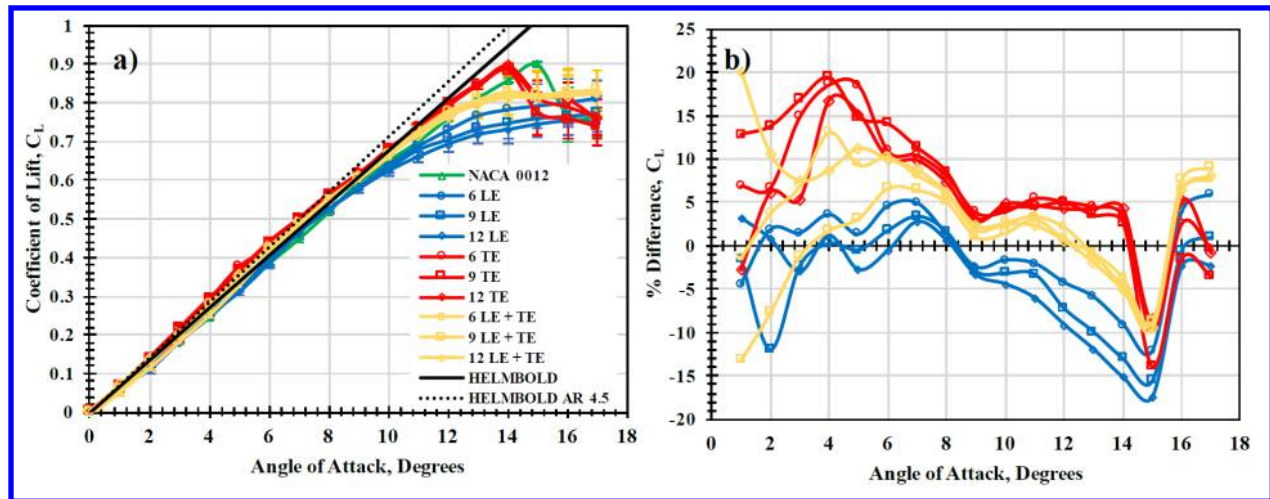


Fig. 9 Variation of C_L a) comparison of all cases and b) percent difference from baseline NACA 0012 wing.

1. Sensitivity on Undulation Placement

Wings with LEU are illustrated in Fig. 9 by blue curves. The lift coefficient for the LEU wings is comparatively lower than all the other cases, especially at higher angles of attack. However, no discernible stall angle was observed and the lift coefficient continues to increase with the increase in angle of attack and does not reach its maximum value within the angle of attack range considered. Similar post-stall behavior in the lift coefficient was documented in literature investigating the wings with leading-edge undulations [14], [16], [17]. At 17° , the LEU wing C_L exceeds that of TEU wing C_L . The differences between LEU wing lift coefficient and the baseline are shown in Fig. 9b. Up to 8° angle of attack, the LEU wing lift coefficient is similar to the baseline but after 8° , the LEU wing lift coefficient decreases significantly on an average of 10%, the highest decrease being 17% at 16° . However, at 17° , the wing starts to perform better than the baseline as the LEU wing C_L continues to increase while the baseline wing C_L decreases post-stall.

TEU wings are represented by red curves in Fig. 9. The TEU wings gave better C_L production out of all other undulated wings, even outperforming the baseline until the stall angle. Unlike the LEU wings, the TEU wings show an abrupt stall at 14° . Fig. 9b illustrates that the TEU wings produce more lift until 14° angle of attack outperforming the baseline by an average of 10%. However, at angles of attack greater than 15° , the performance of the TEU wings is similar to that of the baseline.

LETEU wings are illustrated by yellow curves in Fig. 9. The C_L trend for this case mirrors to that of both the LEU wing and TEU wing with no discernible stall angle and higher C_L than baseline wing at low angles of attack. The LETEU wings follow very closely with the TE wings up until the wings begin to stall at 12° at which point C_L begins to deviate. Although C_L is lower for LETEU wings as compared to the TEU wings from the region of 12° to 15° , C_L continues to increase and shows superior performance at 17° angle of attack. The better overall performance of the LETEU wing cases is also documented in similar studies where the undulations in leading and trailing edges are investigated, especially at the wingtips [15].

2. Sensitivity on Number of Undulations

The LEU wing cases in Fig. 9b show that from 3° to 15° , the wing with 6 undulations exhibits a higher C_L at higher angles of attack. At lower angles of attack, there are no significant differences between LEU wings with different numbers of undulations. The TEU and LETEU wings however, the differences in the C_L are scattered at lower angles of attack and from 9° , the C_L becomes independent of the number of undulations. From these observations, it appears there are no strong correlations between wing performance and the number of undulations along the wingspan, especially before stall.

3. Variation in Lift Curve Slope

The lift curve slope of each case was calculated using the linear regions of Fig. 9a and is represented in Fig.

10. The experimental baseline NACA 0012 lift curve slope of 0.0648 deg^{-1} is shown as a dotted line in Fig. 10 for better comparison. The LEU wing lift curve slope is similar to that of the baseline and the TEU and LETEU wing cases shows an increment in lift curve slope by 5%. This slight increment in lift curve slope could be attributed to the slight increase in the effective aspect ratio of the undulated wings. A different trend is observed for the TEU wings where the lift curve slope decreases from 6 to 9 undulations and then increases from 9 to 12 undulations while the other cases continually decrease in the lift curve slope with increase in number of undulations. Despite the trends, the change in lift curve slope magnitude does not exceed 0.001 deg^{-1} within any undulation placement group, which suggests that the number of undulations does not have a significant impact on the wing performance. However, in terms of the undulation locations, the same trends are observed in Fig. 10 as those observed in Fig. 9 as the TEU wings show the highest C_L and lift curve slope, the LEU wings perform least favorably in both C_L and lift curve slope, and the LETEU wings perform better than LEU wings but not as well as TEU wings.

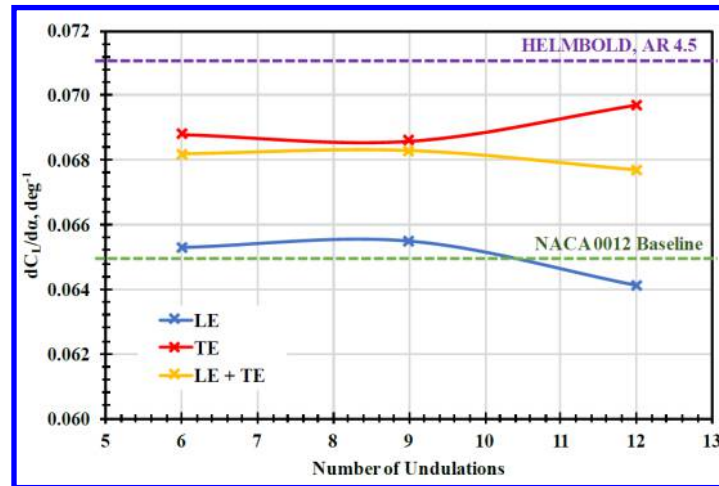


Fig. 10 Variation of lift curve slope with undulation placement and number of undulations

B. Coefficient of Drag

The coefficient of drag results from the force-based experiments are shown in Fig. 11a. The variation in C_D with angle of attack for each undulated wing is compared to the baseline NACA 0012 wing in terms of percent difference in Fig. 11b.

1. Sensitivity on Undulation Placement

Similar to lift coefficient variation, there are no significant differences between the baseline and the other undulation cases until 8° angle of attack. However, after 8° , significant deviations from the baseline case are observed. The drag coefficient of the LEU wings is significantly greater than the baseline at higher angles of attack as observed in Fig 11b. However, as observed in the lift coefficient variation, there are no abrupt increase in coefficient of drag due to stall which is beneficial for maneuverability. On average, the LEU wing increases drag by 30% when compared to the baseline at angles of attack lower than 15° . At 17° , the percent difference in drag coefficient decreases, a trend observed in all the undulated wing cases.

The TEU wings showed an overall favorable drag performance as compared to the baseline and the LEU wings, especially in the range of 8° to 14° angles of attack as observed in Fig 11b. In this range, the 12 TEU wing achieved a reduction in C_D of approximately 10% as compared to the NACA 0012 wing. However, similar to the lift coefficient variation, the onset of stall is marked by a sudden increase in drag coefficient. And soon after the stall angle of attack, there is an abrupt reduction in drag coefficient at 17° as observed in Fig 11b.

Similar to the lift coefficient case, the LETEU wing performs better at lower angle of attack with comparatively lower drag coefficient and also shows post stall benefits at higher angle of attack. Unlike TEU case, the drag coefficient increases gradually near the stall angle of attack. As shown in Fig. 11b, the percent difference of all LETEU wing cases are favorable from the range of 8 to 11° , where the percent difference is similar to that of the TEU wings.

2. Sensitivity on Number of Undulations

While the lift coefficient did not show any significant differences as a function of number of undulations, noticeable differences are seen in the drag coefficient, as seen in Fig 11b. The LEU wing cases show high scatter in the percentage difference in the drag coefficient from the baseline for different numbers of undulations, especially at lower angles of attack. The scatter in the coefficient of drag deviation from the baseline is lower for the LETEU case. However, no discernable trends in the number of undulations are observed between the different undulation placements cases.

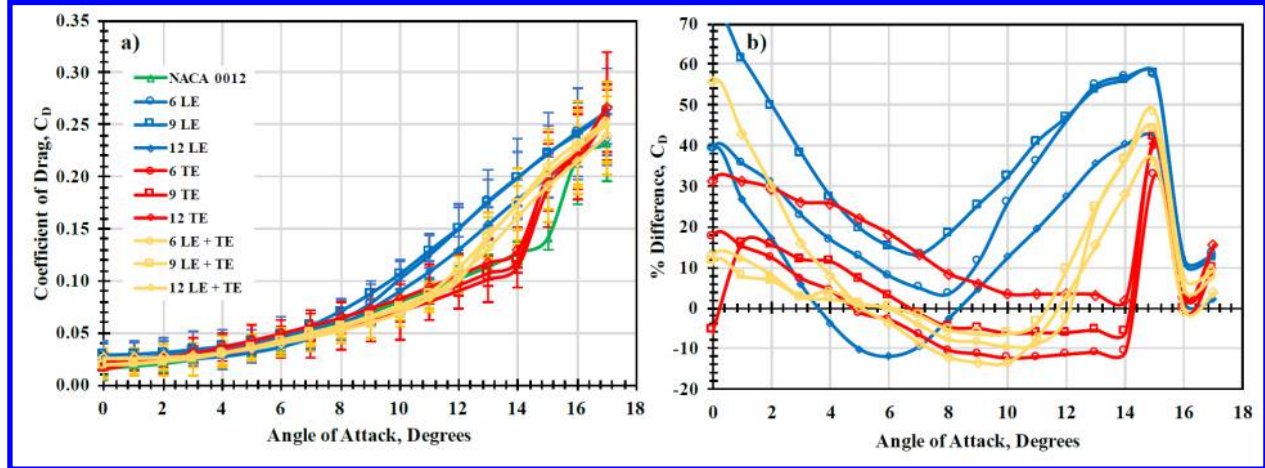


Fig. 11 Variation of C_D a) comparison of all cases and b) percent difference from baseline NACA 0012 wing.

3. Induced Drag and Parasite Drag

The lift coefficient variation for different cases shown in Fig 9 was used to determine the coefficient of induced drag using Eq. (2) below

$$C_{Di} = \frac{C_L^2}{\pi e AR} \quad (2)$$

where e is the span efficiency which was assumed to be 0.7 for all the wing cases. The caveat here is that the span efficiency for the different undulated wings will not likely be the same. However, deviations in the span efficiency within the different undulated cases are assumed to be smaller. The induced drag coefficient is then subtracted from the total drag coefficient obtained from the experiment to determine the parasitic drag coefficient. The variation of induced and parasitic drag coefficient is shown in Fig. 12a and 12b, respectively. All the undulated wing cases show lower induced drag when compared to the baseline due to the lower C_L produced in some cases and due to the increased effective aspect ratio of the undulated cases from 4 to 4.5. Due to lower lift production, the LEU wing cases resulted in a lower induced drag coefficient when compared to the baseline and other wing cases, followed by the LETEU case. The TEU case however, shows similar induced drag coefficient magnitude when compared to the baseline even though higher C_L was observed for the TEU case.

The parasitic drag coefficient provides key insight into the way the contours are affecting the profile and pressure drag of the wing. At lower angle of attack cases, the parasitic drag stays relatively constant with a slight non-linear increase for wing cases. At an angle of attack of 7° , the LEU case shows significant increase in the parasitic drag coefficient whereas the TEU wing case shows almost constant parasitic drag coefficient, similar to the baseline case until stall. As usual, the LETEU wing case falls within the LEU and TEU case.

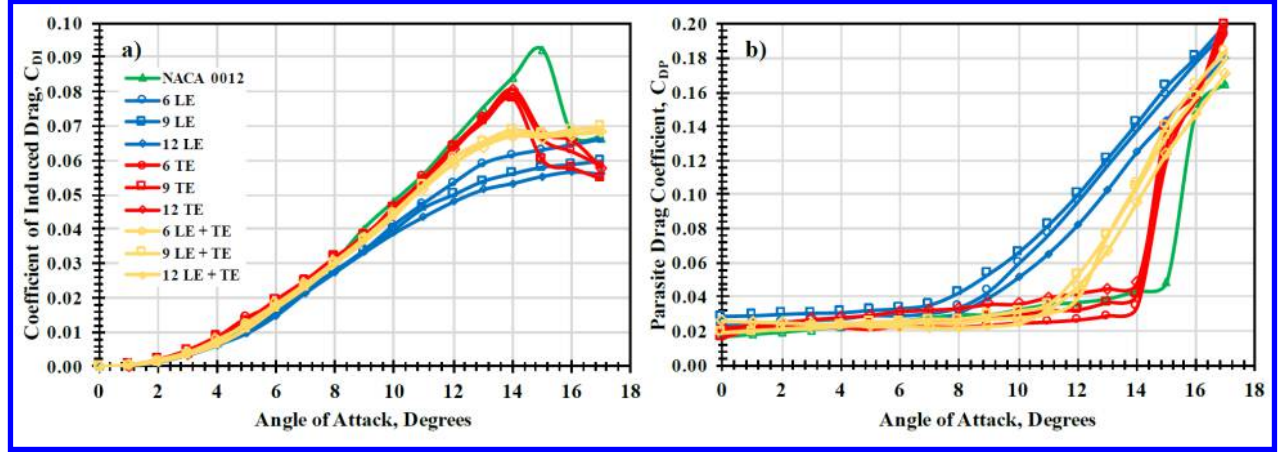


Fig. 12 Variation of a) induced drag and b) parasite drag with angle of attack

C. Aerodynamic Efficiency

The aerodynamic efficiency quantified by the ratio of C_L and C_D for the different wing cases are shown in Fig. 13a along with the percentage differences in the efficiency between the undulated wing cases and the baseline shown in Fig. 13b.

1. Sensitivity on Undulation Placement

As expected, the LEU wings show the least favorable aerodynamic efficiency magnitude and variation with respect to angle of attack when compared to other wing cases. The LEU wings exhibited a lower aerodynamic efficiency than the baseline wing at all angles of attack with the exception of 12 LEU wing from 4 to 8°. The C_L/C_D ratio reaches its peak from 6 to 7° for the LEU wings and then decreases with increase in angle of attack falling significantly less than the baseline wing efficiency. As shown in Fig. 13b, the LEU wings show a significant reduction in aerodynamic efficiency beyond 7° angle of attack. To analyze the way the leading edge undulations affect the wingtip vortex and free shear layer, the 12 LEU wing was considered to be investigated through PIV technique.

The TEU wings on the other hand showed an overall favorable aerodynamic efficiency compared to the baseline wing with the exception of 12 TE which shows a trend similar to that of the baseline. These variations were to be expected as the TEU wings produce higher lift and lower drag than the baseline at most of the angle of attack range considered. Fig. 13b shows that the 6 TE case achieves almost 20% increase in aerodynamic efficiency as compared to the baseline wing and does it consistently from 6° to 14°. To analyze the way the trailing edge undulations affect the wingtip vortex and free shear layer, the 6 TEU wing was considered to be investigated through PIV technique.

All LETEU wings showed a higher C_L/C_D than the baseline from the range of 4 to almost 12°, and then decrease in a trend similar to that of the LEU wings when the wings reach stall. Fig. 13b indicates that the 12 LETEU wing achieved an increase in aerodynamic efficiency of almost 20% from approximately 6 to 8°. To analyze the way including both leading edge and trailing edge undulations affect the wingtip vortex and free shear layer, the 12 LETEU case will represent the LETEU wings during PIV experiments.

2. Sensitivity on Number of Undulations

While the 6 TEU case performs the best out of the TEU cases, the 12 LEU and 12 LETEU cases perform best out of the LEU and LETEU cases. This shows that there is no strong correlation between aerodynamic efficiency and the number of undulations. The maximum deviation in the aerodynamic efficiency between the number of undulations for any particular case is on the order of 10% with greater deviations seen in both the LEU and TEU cases as compared to the LETEU cases. This could be attributed to the scatter in the C_D measurements at lower angles of attack. All the LETEU wings showed a higher C_L/C_D than the baseline from the range of 4° to almost 12° angle of attack, and then decrease in a trend similar to that of the LEU wings at stall. Fig. 13b indicates that the 12 LETEU wing achieved an increase in aerodynamic efficiency of almost 20% from approximately 6 to 8°. To analyze the way both leading edge and trailing edge undulations affect the wingtip vortex, high performing cases in each undulation

placement category: the 12 LE TEU case, 6 TEU case and 12 LEU case was investigated through PIV.

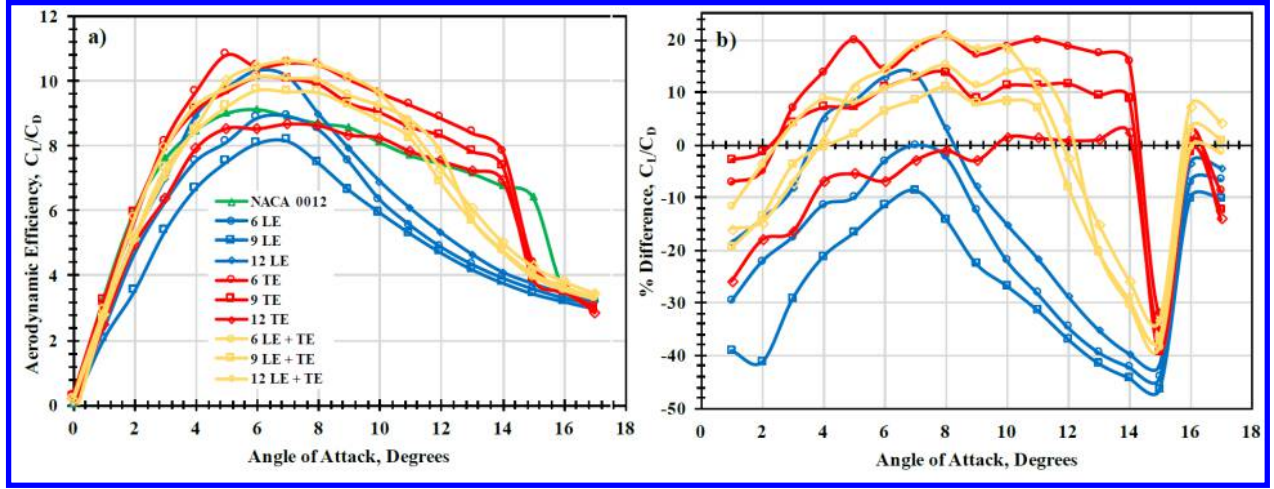


Fig. 13 Variation of C_l/C_d a) comparison of all cases and b) percent difference from NACA 0012 wing.

IV. Wingtip Vortex PIV Results

A. Vortex Wandering Corrections

The oscillations of the wingtip vortex in the cross-stream plane, referred to as vortex wandering, if not corrected for, biases the wingtip vortex measurements causing the vortex to have a large radius with lower vorticity and circulation than it does. The vortex wandering was quantified by tracking the vortex center across the PIV image using Q-criterion, a well-established method to find vortex center in the flow. The equation for Q-criterion is

$$Q = \frac{1}{2} (\|\Omega_{ij}\|^2 - \|R_{ij}\|^2) \quad (3)$$

Where Ω_{ij} is the rotation tensor given by

$$\Omega_{ij} = \frac{1}{2} \begin{bmatrix} 0 & \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}\right) \\ \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) & 0 \end{bmatrix} \quad (4)$$

and S_{ij} is the strain tensor given by

$$S_{ij} = \frac{1}{2} \begin{bmatrix} 2\frac{\partial u}{\partial x} & \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \\ \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) & 2\frac{\partial v}{\partial y} \end{bmatrix} \quad (5)$$

An example Q-criterion contour is shown in Fig. 14 where a higher Q magnitude is observed in the vortex center indicating a larger vorticity magnitude and lower strain magnitude in the vortex center. The peak Q-locations in each image pairs were found and shifted to the center of the field of view. Once the vortex centers were determined, the RMS wandering amplitude σ given by Eq. (6) was quantified

$$\sigma = \sqrt{\sigma_x^2 + \sigma_y^2} \quad (6)$$

where σ_x is the RMS wandering amplitude component in the horizontal x-direction and σ_y is the RMS wandering amplitude component in the vertical y-direction. On average the σ in each case was determined to be around 0.014 meters.

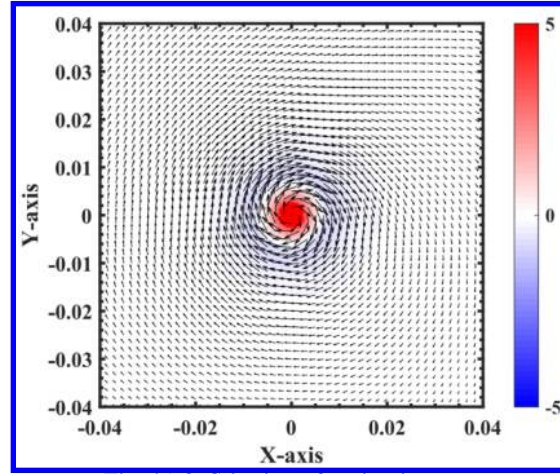


Fig. 14 Q-Criterion of a wingtip vortex

B. Wingtip Vortex Azimuthal Velocity

In order to quantify the effect of undulations on the wingtip vortex growth and evolution, the wingtip vortex normalized azimuthal velocity profiles of the baseline and the undulations wings are compared against the well-known Batchelor's wingtip vortex model given by

$$\frac{U}{U_{Max}} = \left(1 + \frac{1}{2\alpha_L}\right) \frac{1}{\eta} (1 - \exp(-\alpha_L \eta^2)) \quad (7)$$

where U_{Max} is the maximum velocity in the X direction, α_L is Lambs constant: 1.256, η is r/r_c where r is the radial location in the wingtip vortex and r_c is wingtip vortex core radius determined by the radial location where maximum azimuthal velocity occurs.

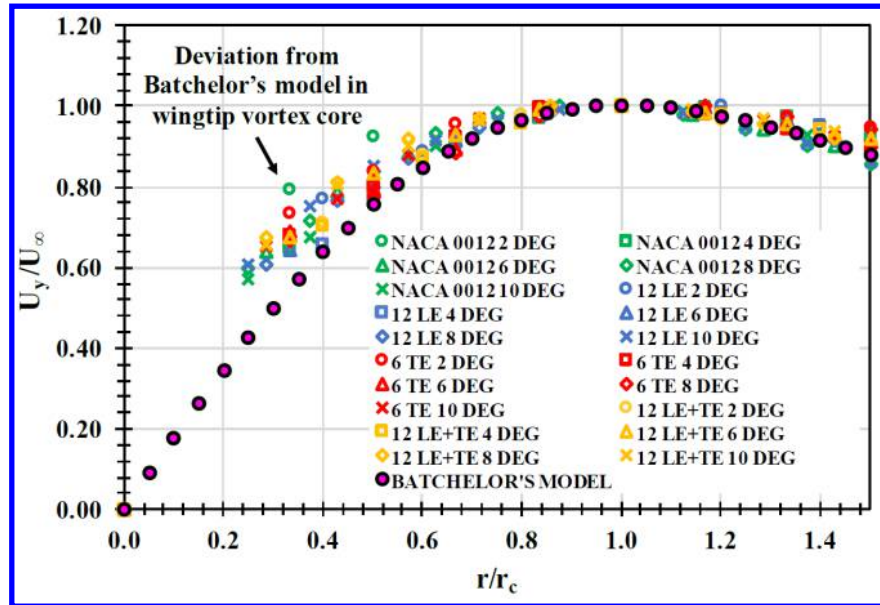


Fig. 15 Variation in normalized u-velocity with respect to radial location

For the majority of the cases shown in Fig. 15, it can be seen that the velocity distribution follows the Batchelor's model trend very well at r/r_c greater than 1. However, inside the wingtip vortex core, deviations from the Batchelor's model can be observed, especially at lower angles of attack. This due to the non-mature status of the wingtip vortex at lower angle of attack cases and similar behavior is documented abundantly in literature. The wings with undulations also show similar behavior as the baseline but doesn't show higher deviations from the model at lower angles of attack

(2°) as observed in the baseline case.

C. Wingtip Vortex Vorticity

The normalized z-vorticity of the wingtip vortex was determined using Eq. (4) for the baseline and undulated wings (Fig. 16). It can be seen that the magnitude of the vorticity increases with an increase in angle of attack for all cases considered. A quick visual investigation of the contours reveal no significant variations between the vorticity contours with the exception of 12 LEU wing case. The wingtip vortex was found to be smaller at each angle of attack when compared to the other cases. This indirectly indicates a reduction in lift as observed in the lift coefficient plot shown in the previous section.

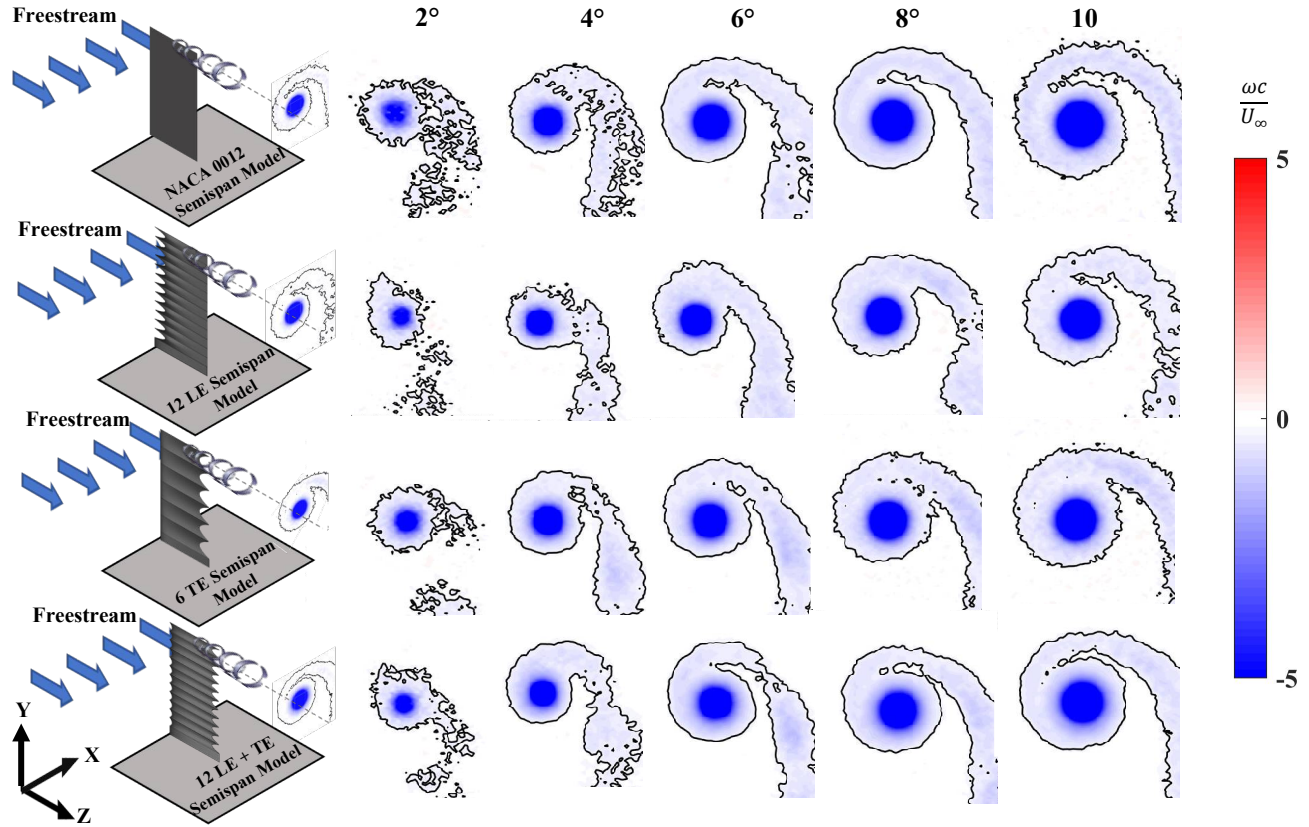


Fig. 16 Normalized vorticity from 2 to 10 degrees angle of attack for baseline and undulated wings

D. Wingtip Vortex Circulation

To quantify the reduction in wingtip vortex strength, the circulation of the wingtip vortex was obtained by integrating the vorticity as shown in Eq. (8).

$$\Gamma = \iint \Omega \cdot ds \quad (8)$$

where Γ is the circulation, Ω is vorticity, and ds is the incremental surface area. After determining the vortex center through Q-criterion, the wingtip vortex circulation was determined as a function of wingtip vortex radius. This wingtip vortex circulation as a function of vortex radius obtained from the experimental data is then compared to the ideal Lamb-Oseen vortex model described by Eq. (9)

$$\Gamma(r) = \Gamma_0 \left(1 - \exp \left(-\frac{r^2}{r_c^2} \right) \right) \quad (9)$$

The Γ_0 in Eq. (9) was changed until a good fit was achieved between the model and the experimental data. This method of determining wingtip vortex circulation was developed by Stevens [24] and has been employed in Corkery et al. [25] and Stevens and Babinsky [26]. Figure 17 shows an example of experimental data compared to the Lamb-Oseen vortex to find wingtip vortex circulation. The Lamb-Oseen model is fitted with the experimental data for the 12 LEU wing at 10° angle of attack by finding the Γ_0 which results in the highest R^2 value. In this scenario, a value of $0.6 \text{ m}^2/\text{s}$ resulted in the highest R^2 value of 0.999. The same method was used to obtain the circulation for all wings from 2 to 10° . The average R^2 value for all cases was 0.985.

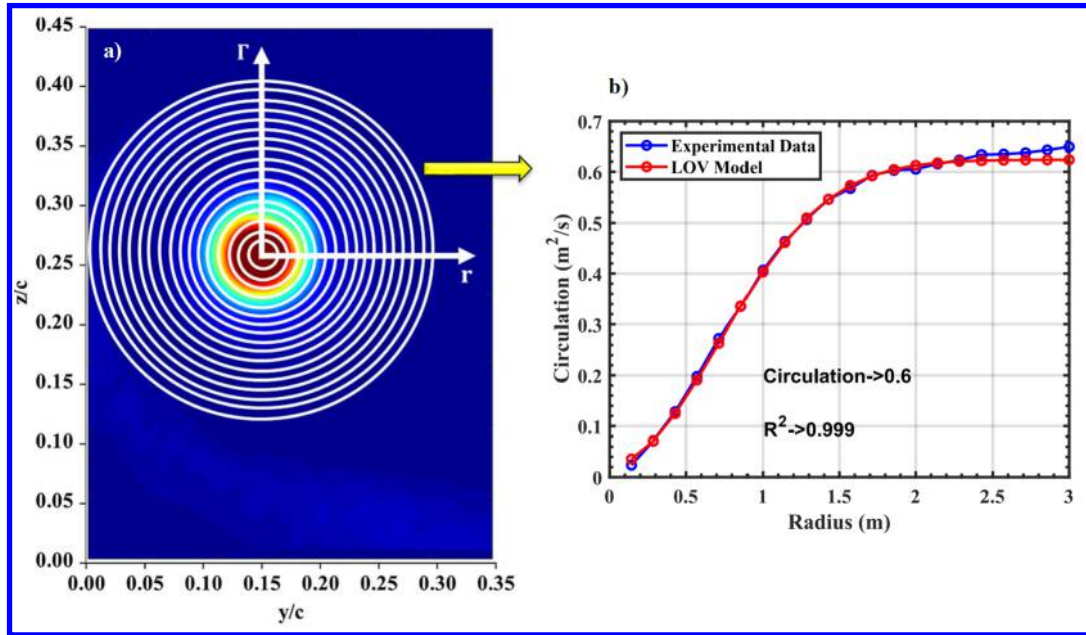


Fig. 17 a) Variation of circulation as a function of vortex radius b) Determining total circulation of wingtip vortex using the Lamb-Oseen vortex model

The variation of normalized circulation with angle of attack for each case is shown in Fig. 18a. For all cases, circulation increases with increase in angle of attack as expected. The undulated wings show similar variation compared to the baseline at 2° and 4° before beginning to trend below the baseline from 6 to 10° . This shows that for a given angle of attack, the undulated wings produce a lower circulation, and therefore weaker wingtip vortex than the baseline case. The TEU case shows a higher circulation for a given angle of attack compared to the LEU and LETEU cases. The variation of normalized circulation with $C_{D,i}$ obtained from experimental data is shown in Fig. 18b. The undulated wings trend below the baseline case as $C_{D,i}$ increases. For a given coefficient of induced drag, the undulated wings produce less circulation than the baseline case, with the LETEU cases showing the lowest circulation. This trend is mirrored in the variation of circulation with C_L shown in Fig. 18c. Again, the undulated wings trend below the baseline. For the same value of C_L , the undulated wings produce lower circulation than the NACA 0012 wing with as high as almost 33% difference in the LETEU case at a C_L of about 0.64. This shows that the contoured wings are producing the same C_L with a lower circulation. This violates the Kutta-Joukowski theorem of linear variation between the C_L and circulation. However, it should be noted that the Kutta-Joukowski theorem was derived under the assumptions of inviscid and irrotational flow. It is possible that the undulations increased the viscous interactions of the wingtip vortex when compared to the baseline leading to a lower wingtip vortex circulation value for the lift coefficient condition. The change in normalized circulation with C_D is also shown in Fig. 18d. The undulated wings and NACA 0012 wing trend together with the exception of the LEU case. Given a certain circulation value, the LEU case produces a higher C_D than the NACA 0012 wing, TEU case, and LETEU case. This trend is in excellent agreement with the increase in C_D is observed for the LEU case in the force-based data (Fig. 11).

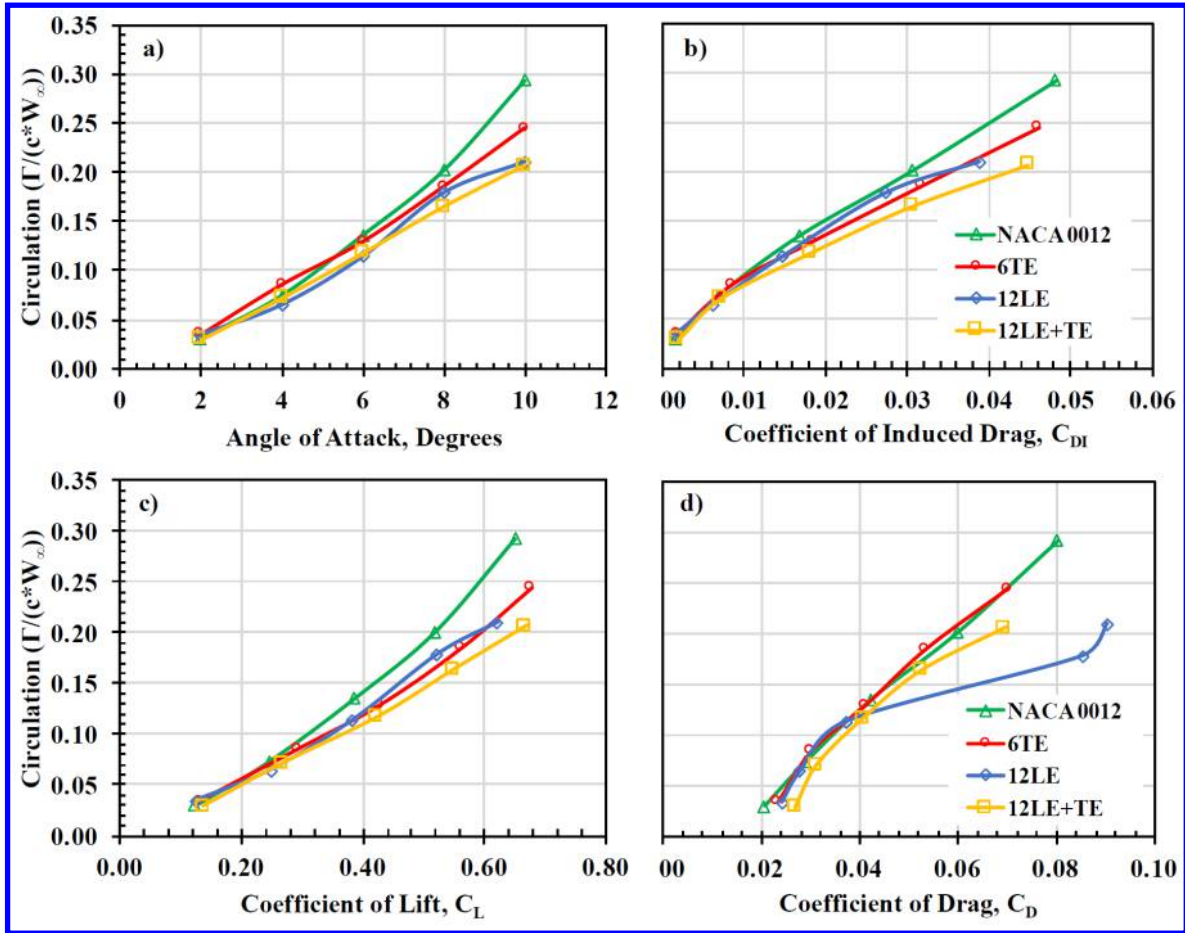


Fig. 18 Variation of circulation as a function of a) angle of attack, b) coefficient of induced drag, c) coefficient of lift, and d) coefficient of drag

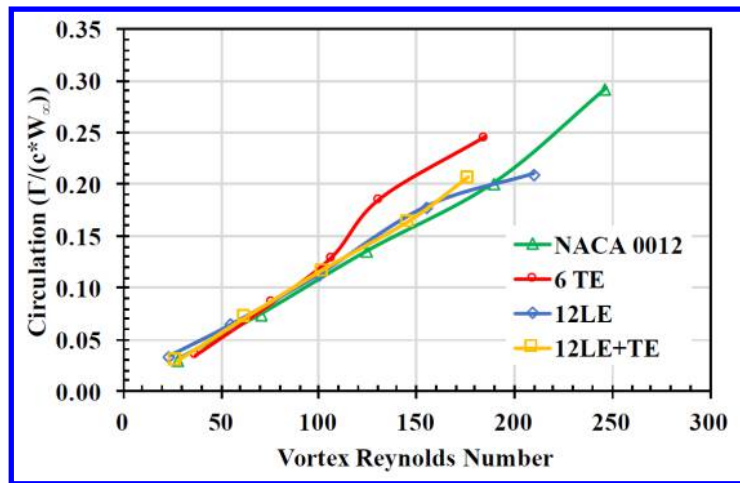


Fig. 19 Variation of wingtip vortex Circulation with wingtip vortex Reynolds number

In order to understand the lower circulation magnitude for the same C_L condition in the undulated wing cases, the variation of wingtip vortex circulation is compared with the wingtip vortex Reynolds number defined by Eq. (10).

$$Re_{vortex} = \frac{\rho V_z r_c}{\mu} \quad (10)$$

where V_z is the wingtip vortex azimuthal velocity, r_c is the wingtip vortex core radius and μ is dynamic viscosity. The vortex Reynolds number indicates the ratio of the vortex inertia to the viscous forces. The circulation of the wingtip vortex for all the cases when plotted against the vortex Reynolds number shows a slight increase in the circulation for a given vortex Reynolds number for the 12 LEU and LETEU cases when compared to the baseline at higher angles of attack. The curves collapse onto one another at lower angles of attack. Significant deviations from the trend is observed after a vortex Reynolds number of 100 for the 6 TEU case. This trend indicates that the higher vortex circulation is achieved for the same vortex Reynolds number when compared to the other cases. This behavior explains the increased lift coefficient for the TEU case and the minimum decrement in the induced drag coefficient. Figure 19 also indicates that the balance between the vortex inertia and the viscous forces is affected by the use of undulations.

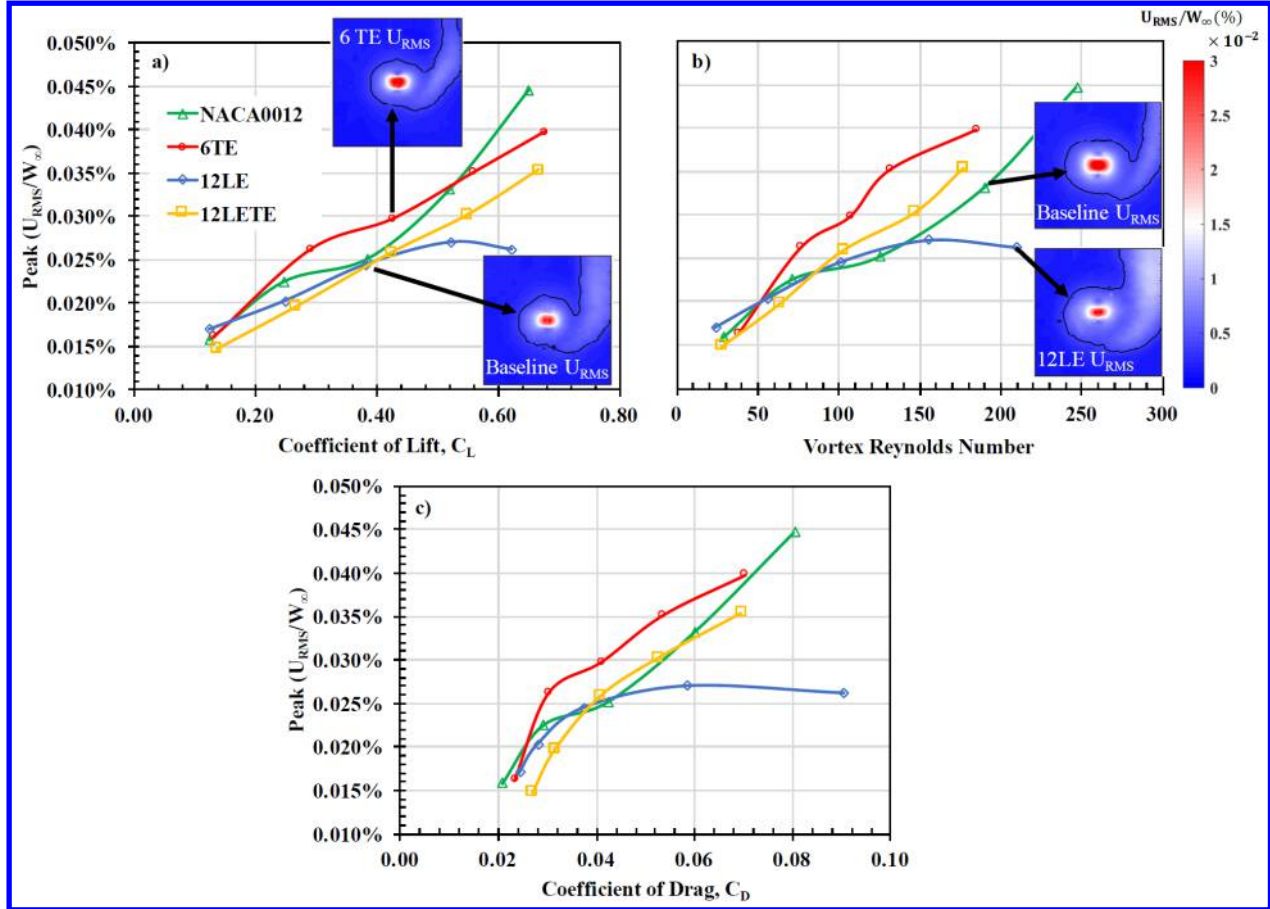


Fig. 20 Variation of the normalized peak U_{RMS} as a function of a) C_L b) vortex Reynolds number and c) C_D

The decrement in the wingtip vortex Reynolds number is compensated by the increase in the wingtip vortex RMS of the velocity fluctuations. The RMS of the velocity is calculated by

$$U_{RMS} = \sqrt{u'^2} \quad (11)$$

where u' is the fluctuating velocity about the x-axis. The variation of the freestream normalized peak U_{RMS} with coefficient of lift, the vortex Reynolds number and the coefficient of drag is shown in Fig. 20a, 20b and 20c respectively. It is readily observed that the 6 TEU wing case has the highest RMS until a certain lift coefficient of value of 0.4. As the C_L increases, the RMS in the baseline case increases. The contour plots of the U_{RMS} around the C_L value of 0.4 is shown in Fig. 20a. for both the 6TEU wing case and the baseline. The contour plots indicate the trend observed in the peak U_{RMS} . The 6 TEU case has the highest RMS in the core of the vortex when compared to the baseline for the same value of C_L . Similar trends are also observed in the variation of peak RMS with the vortex

Reynolds number. The TEU wing case has the higher RMS for a given vortex Reynolds number indicating the increased turbulence level in the vortex when compared to the other cases. There seem to be a correlation between the RMS and the aerodynamic performance of the wing. At lower angles of attack, as indicated in the lift, drag and aerodynamic coefficient plots, the TEU wing case delivered a superior performance when compared to the baseline. And correspondingly, the RMS in the vortex core for the 6 TEU wing case is also higher than the other cases for a given C_L and the vortex Reynolds number. This correlation is still observed on the other end of the spectrum. The 12 LEU wing case delivered poor performance overall at lower angles of attack as shown in the lift, drag coefficient and the aerodynamic performance plots. Coincidentally, the wingtip vortex core RMS is significantly lower at any given C_L condition greater than 0.4 and a given vortex Reynolds number greater than 100. Similar trend is also seen in the peak RMS vs with C_D where the 6TEU wing shows greater RMS at a given coefficient of drag condition when compared to all the other cases. Coincidentally, the 12LEU wing case shows lower RMS when compared at a given C_D condition. All these correlations indicate that the vortex RMS is a good indicator of the aerodynamic performance of the wing.

IV. Conclusions

Wings with airfoil preserved undulations were investigated to determine its effect on the aerodynamic performance and on the balance of induced and parasite drag of the wing.

- The force coefficients data shows clear differences in aerodynamic performance between the TEU, LEU, and LETEU cases even at lower angles of attack. The TEU wing cases performs better than the LEU and LETEU wing cases, especially at lower angles of attack. At higher angles of attack however, the LEU and LETEU wing cases hint the post-stall benefits observed in the previous literature. It was observed that the changes between the TEU, LEU and LETEU cases were due to the way the wings change the balance of induced and parasite drag.
- The lift coefficient seems to be independent of the number of undulations on the TEU, LEU and LETEU wing cases. However, subtle changes were observed between the number of undulations in drag coefficient and in the aerodynamic performance. However, no discernible trends were observed.
- The top performing candidates of each category: 6 TEU, 12 LEU and 12 LETEU cases were chosen for PIV investigations, to provide insight to the way it affects the balance of induced and parasite drag of the wing. The 6 TEU wing case, providing superior aerodynamic efficiency (higher lift coefficient but lower induced drag) showed an increased circulation for a similar vortex Reynolds number when compared to the other cases. The reduction in the wingtip vortex Reynolds number is compensated by the increased RMS in the wingtip vortex core. This relationship is also observed in the LEU wing case which showed poor aerodynamic performance at low angles of attack and resulting in a lower RMS in the wingtip vortex core. This relationship indicate that vortex RMS has an inverse relationship with the aerodynamic performance and the wingtip vortex RMS plays a key role in affecting the balance of induced and parasite drag of the wing.

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